Exercise 5.8

Is C_2 uniquely decodable?

Solution:

Yes, because we can determine all the positions where 101 is from where 0s are. Then, we can determine that the remaining 1s are coded from 1s. The codewords can be uniquely decomposed and decoded.

Exercise 5.14

Prove the result stated above, that for any set of codeword lengths $\{l_i\}$ satisfying the Kraft inequality, there is a prefix code having those lengths.

Solution:

Sort l_i in descending order, and fill them into the budget diagram from top to bottom.
When the Kraft inequality is satisfied, there will always be a fit without any overlaps i.e. a prefix code.

Exercise 5.16

Prove that there is no better symbol code for a source than the Huffman code.

Solution:

Sibling lemma:
In an optimal code, if $p_i < p_j$ then $l_i \geq l_j$ since otherwise we can swap p for i and j to strictly decrease L.
Also, two longest codewords must be equal length (otherwise truncate) and can be taken to be siblings (differ by only last bit).
These prove that two least-probable symbols can be constructed to be siblings at max depth.

Now, we prove that Huffman coding is optimal by induction on n (the number of symbols):

(i) Base: Optimal $L = 1$, which is what Huffman constructs.

(ii) Inductive step:
Assume Huffman coding is optimal for $n-1$ symbols. Let a, b be the least probable symbols of n symbols.
Huffman merges a, b into a new symbol and applies the algorithm to the remaining $n-1$ symbols (optimal by hypothesis with length L'), then

splits the new symbol into a and b. This means Huffman's length is $L_H = L' + p_a + p_b$.
Now, assume $L_{opt} \leq L_H$. By the lemma we can assume a, b are siblings and by merging we get a symbol code of $L_{opt} - p_a - p_b$. Since L' optimal, $L_{opt} - p_a - p_b \geq L'$.
Thus $L_{opt} \geq L' + p_a + p_b = L_H \geq L_{opt}$ so $L_H = L_{opt}$. Hence Huffman coding optimal for n symbols.

Exercise 5.19

Is the code $\{00, 11, 0101, 111, 1010, 100100, 0110\}$ uniquely decodable?

Solution:

No, since '111111' can be decoded as '11/11/11' or '111/111'.

Exercise 5.20

Is the ternary code $\{00, 012, 0110, 0112, 100, 201, 212, 22\}$ uniquely decodable?

Solution:

This is a prefix code so yes, it is uniquely decodable.

Exercise 5.21

Make Huffman codes for X^2, X^3 and X^4 where $\mathcal{A}_X = \{0, 1\}$ and $\mathcal{P}_X = \{0.9, 0.1\}$. Compute their expected lengths and compare them with the entropies $H(X^2)$, $H(X^3)$ and $H(X^4)$. Repeat this exercise for X^2 and X^4 where $\mathcal{P}_X = \{0.6, 0.4\}$.

Solution:

X^2

00	0.81	0
01	0.09	10
10	0.09	110
11	0.01	111

$L(C, X) = 0.81 \times 1 + 0.09 \times 2 + 0.1 \times 3 = 1.29$
$H(X^2) = 0.9380$

X^3

000	0.729	0
001	0.081	100
010	0.081	101
100	0.081	110
011	0.009	11100
101	0.009	11101
110	0.009	11110
111	0.001	11111

We get $L(C, X) = 1.598$, $H(X^3) = 1.407$

X^4

0000	0.6561	0
0001	0.0729	100
0010	0.0729	101
0100	0.0729	110
1000	0.0729	1110
0011	0.0081	111000
0101	0.0081	111001
0110	0.0081	1111010
1001	0.0081	1111011
1010	0.0081	111110
1100	0.0081	1111110
0111	0.0009	11111100
1011	0.0009	11111101
1101	0.0009	111111110
1110	0.0009	1111111110
1111	0.0001	1111111111

We get $L(C, X) = 1.9702$, $H(X^4) = 1.876$

Now, for $\mathcal{P}_X = \{0.6, 0.4\}$, $H(X) = 0.9710$ so $H(X^2) = 1.9420$ and $H(X^4) = 3.8840$.
With a similar procedure as above, we obtain $L(C, X) = 2$ and $L(C, X) = 3.9248$ respectively.

Exercise 5.22

Find a probability distribution $\{p_1, p_2, p_3, p_4\}$ such that there are *two* optimal codes that assign different lengths $\{l_i\}$ to the four symbols.

Solution:

$\{\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6}\}$ is one such distribution.
$l = \{00, 00, 01, 11\}$ and $l = \{0, 10, 110, 111\}$ have both $L = 2$.

Exercise 5.23

(Continuation of exercise 5.22.) Assume that the four probabilities $\{p_1, p_2, p_3, p_4\}$ are ordered such that $p_1 \geq p_2 \geq p_3 \geq p_4 \geq 0$. Let Q be the set of all probability vectors $\mathbf{p}$ such that there are *two* optimal codes with different lengths. Give a complete description of Q. Find three probability vectors $\mathbf{q}^{(1)}, \mathbf{q}^{(2)}, \mathbf{q}^{(3)}$, which are the convex hull of Q, i.e., such that any $\mathbf{p} \in Q$ can be written as

$$\mathbf{p} = \mu_1 \mathbf{q}^{(1)} + \mu_2 \mathbf{q}^{(2)} + \mu_3 \mathbf{q}^{(3)}, \tag{5.29}$$

where $\{\mu_i\}$ are positive.

Solution:

For there to be two optimal codes with different lengths, since there are 3 merges taken, there must be 2 different ways to merge in the second merge (For first merge, it just makes it symmetric, for last, there is only 1 possible merge).
For that, p_1 must be equal to $p_3 + p_4$. Together with the other conditions,

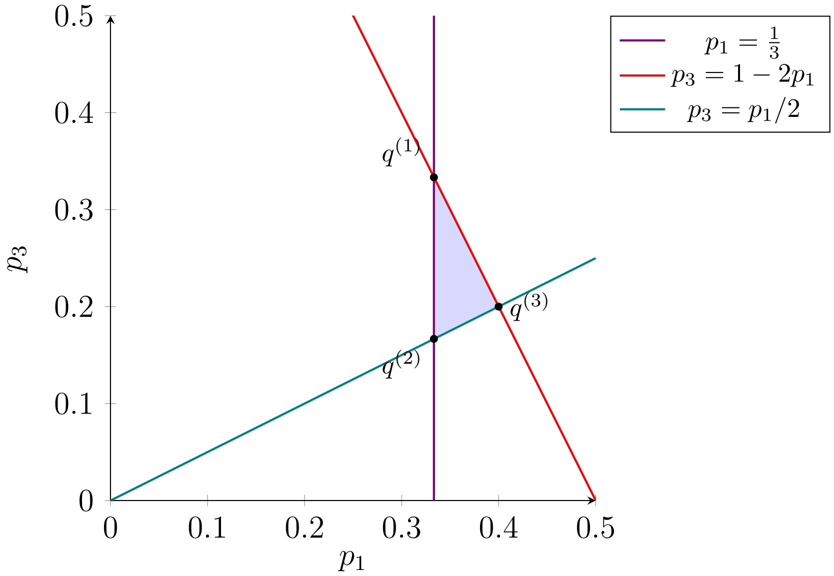
$$\begin{cases} p_1 = p_3 + p_4 \\ p_1 + p_2 + p_3 + p_4 = 1 \\ p_1 \geq p_2 \geq p_3 \geq p_4 \geq 0 \end{cases}$$

$\iff$

$$\begin{cases} p_1 = p_3 + p_4 \\ 2p_1 + p_2 = 1 \\ p_1 \geq p_2 \geq p_3 \geq p_4 \geq 0 \end{cases}$$

$\iff$

$$\begin{cases} p_1 \geq \frac{1}{3} \\ p_3 \leq 1 - 2p_1 \\ p_3 \geq \frac{p_1}{2} \end{cases}$$



The above area shows the possible p_1, p_3 pair (which uniquely determines p_2, p_4 from the conditions).
Therefore, from linearity we get

$$q_1 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0)$$
$$q_2 = (\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$$
$$q_3 = (\frac{2}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5})$$

Exercise 5.24

Write a short essay discussing how to play the game of **twenty questions** optimally. [In twenty questions, one player thinks of an object, and the other player has to guess the object using as few binary questions as possible, preferably fewer than twenty.]

Solution:

To maximise information obtained, we want to bring the estimated probability of yes/no to be close to 50% as possible. This is impractical to do accurately (since considering all possible objects and its likelihood as the answer are impractical), but we can do an approximation by randomly coming up with several objects and think of a question that divides them in 2 (Monte-Carlo approximation, assuming you and the opponent has similar answer distribution).

Exercise 5.25

Show that, if each probability p_i is equal to an integer power of 2 then there exists a source code whose expected length equals the entropy.

Solution:

Set the length for symbol with p_i to be $\log \frac{1}{p_i}$ which is an integer. Then,

$$L(C, X) = \sum_i p_i l_i = \sum_i p_i \log \frac{1}{p_i} = H(X)$$

Exercise 5.26

Make ensembles for which the difference between the entropy and the expected length of the Huffman code is as big as possible.

Solution:

We want to maximise $L(C, X) - H(X) = \sum p_i(l_i - \log \frac{1}{p_i})$.
$l_i - \log \frac{1}{p_i} < 1$ so most contribution will come from the symbol that has large p_i. We want to make $l_i - \log \frac{1}{p_i}$ as close to 1 for such symbols.
Consider two-symbol ensemble with probabilities ϵ and $1 - \epsilon$ ($0 < \epsilon < 1$). Then, $L(C, X) = 1$ regardless of the value of ϵ.
However, $H(X) = H_2(\epsilon) \to 0$ as $\epsilon \to 0$.
Hence the ensemble with an extremely skewed probability has a difference between the entropy $\approxeq 1$, which is as big as it gets.

Exercise 5.27

A source X has an alphabet of eleven characters

$$\{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}, \mathbf{f}, \mathbf{g}, \mathbf{h}, \mathbf{i}, \mathbf{j}, \mathbf{k}\},$$

all of which have equal probability, 1/11.
Find an optimal uniquely decodeable symbol code for this source. How much greater is the expected length of this optimal code than the entropy of X?

Solution:

With Huffman coding,

symbol	codeword	length
a	000	3
b	001	3
c	010	3
d	011	3
e	1000	4
f	1001	4
g	1010	4
h	1011	4
i	1100	4
j	1101	4
k	111	3

$$H(X) = \log_2 11 = 3.459$$
$$L(C, X) = \frac{3 \times 5 + 4 \times 6}{11} = 3.545$$
$$L(C, X) - H(X) = 0.08645$$

Exercise 5.28

Consider the optimal symbol code for an ensemble X with alphabet size I from which all symbols have identical probability $p = 1/I$. I is not a power of 2.
Show that the fraction f^+ of the I symbols that are assigned codelengths equal to

$$l^+ \equiv \lceil \log_2 I \rceil \tag{1}$$

satisfies

$$f^+ = 2 - \frac{2^{l^+}}{I} \tag{2}$$

and that the expected length of the optimal symbol code is

$$L = l^+ - 1 + f^+. \tag{3}$$

By differentiating the excess length $\Delta L \equiv L - H(X)$ with respect to I, show that the excess length is bounded by

$$\Delta L \leq 1 - \frac{\ln(\ln 2)}{\ln 2} - \frac{1}{\ln 2} = 0.086. \tag{4}$$

Solution:

When all symbols have equal probability, the merging process will be the same as if there are $2^{\lceil \log_2 I \rceil}$ symbols (a full tournament) but with $2^{l^+} - I$ dummies and the dummies being merged with their adjacents. The way the symbols are allocated to the tournament slots is from the left slots of the lowest stage, left to right (will fill in all) and then right slots (, left to right). Then, $2^{l+} - I$ symbols will not have their neighbour so will have length $l^+ - 1$ while the remaining will have length l^+. Hence,

$$\begin{aligned} I(1 - f^+) &= 2^{l+} - I \\ If^+ &= 2I - 2^{l+} \\ f^+ &= 2 - \frac{2^{l+}}{I} \end{aligned}$$

Also,

$$\begin{aligned} L &= l^+ f^+ + (l^+ - 1)(1 - f^+) \\ &= l^+ f^+ + l^+ - l^+ f^+ - 1 + f^+ \\ &= l^+ - 1 + f^+ \end{aligned}$$

Now,

$$\begin{aligned} \Delta L &\equiv L - H(X) \\ &= \lceil \log_2 I \rceil - 1 + 2 - \frac{2^{l+}}{I} - \log_2 I \\ &= 1 - \frac{2^{l+}}{I} + \lceil \log_2 I \rceil - \log_2 I \\ \frac{d\Delta L}{dI} &= \frac{2^{l+}}{I^2} + 0 - \frac{1}{I \ln 2} \quad (\because I \text{ not power of } 2) \end{aligned}$$

$$\begin{aligned} &\frac{d\Delta L}{dI} = 0 \\ \iff & 2^{l+} - \frac{I}{\ln 2} = 0 \\ \iff & I = 2^{l+} \ln 2 \end{aligned}$$

To confirm this critical point is a maximum, consider the second derivative:

$$\frac{d^2 \Delta L}{dI^2} = -\frac{2 \cdot 2^{l+}}{I^3} + \frac{1}{I^2 \ln 2}$$

Evaluating at $I = 2^{l+} \ln 2$,

$$\begin{aligned} \left.\frac{d^2 \Delta L}{dI^2}\right|_{I = 2^{l+} \ln 2} &= \frac{1}{I^2} \left(\frac{1}{\ln 2} - \frac{2 \cdot 2^{l+}}{I} \right) \\ &= \frac{1}{I^2} \left(\frac{1}{\ln 2} - \frac{2 \cdot 2^{l+}}{2^{l+} \ln 2} \right) \\ &= \frac{1}{I^2} \left(\frac{1}{\ln 2} - \frac{2}{\ln 2} \right) = -\frac{1}{I^2 \ln 2} < 0, \end{aligned}$$

so ΔL is concave at this point, confirming it is a (local, and since it is the only critical point, global) maximum on the interval. So

$$
\begin{aligned}
\Delta L &\le 1 - \frac{1}{\ln 2} + \log_2 \frac{2^{l+}}{I} \\
&= 1 - \frac{1}{\ln 2} - \log_2(\ln 2) \\
&= 1 - \frac{\ln(\ln 2)}{\ln 2} - \frac{1}{\ln 2} \\
&= 0.086
\end{aligned}
$$

Exercise 5.29

Consider a sparse binary source with $\mathcal{P}_X = \{0.99, 0.01\}$. Discuss how Huffman codes could be used to compress this source *efficiently*. Estimate how many codewords your proposed solutions require.

Solution:

Let a be the symbol that has probability 0.99 and b 0.01. Then if we pair two symbols and apply Huffman coding,

string	probability	codeword
aa	0.9801	0
ab	0.0099	10
ba	0.0099	110
bb	0.0001	111

Expected length $= 1\times 0.9801 + 2\times 0.0099 + 3\times 0.0099 + 3\times 0.0001 = 1.0299$. Then, Expected length per source symbol $= 0.51495$ which is much less than pure 1 symbol encoding.
The drawback, however, is that odd length source code cannot be encoded.

Exercise 5.30

Scientific American carried the following puzzle in 1975.

> **The poisoned glass.** 'Mathematicians are curious birds', the police commissioner said to his wife. 'You see, we had all those partly filled glasses lined up in rows on a table in the hotel kitchen. Only one contained poison, and we wanted to know which one before searching that glass for fingerprints. Our lab could test the liquid in each glass, but the tests take time and money, so we wanted to make as few of them as possible by simultaneously testing mixtures of small samples from groups of glasses. The university sent over a mathematics professor to help us. He counted the glasses, smiled and said:
>
> ' "Pick any glass you want, Commissioner. We'll test it first."
>
> ' "But won't that waste a test?" I asked.
>
> ' "No," he said, "it's part of the best procedure. We can test one glass first. It doesn't matter which one." '
>
> 'How many glasses were there to start with?' the commissioner's wife asked.
>
> 'I don't remember. Somewhere between 100 and 200.'
>
> What was the exact number of glasses?

Solve this puzzle and then explain why the professor was in fact wrong and the commissioner was right. What is in fact the optimal procedure for identifying the one poisoned glass? What is the expected waste relative to this optimum if one followed the professor's strategy? Explain the relationship to symbol coding.

Solution:

The exact number of glasses was likely $2^7 + 1 = 129$. The professor's strategy was as follows:
Test one glass, check if the glass is poisoned. Then, for the rest of 2^7, assign a binary number to each and test mixture to each of them that have ith bit 1 from $i = 1$ to 7, and the one that matches will be the poisoned (we can also do by removing half of glasses that are not poisoned each round so we don't have to mix 2^6 glass every time).

$$\mathbb{E}(step) = \frac{1}{129} \times 1 + \frac{128}{129} \times 8 = 7.946$$

The optimal strategy is as follows:
Each has probability of $\frac{1}{129}$ being poisoned, so we could apply Huffman coding to 129 glasses, and similar to above, the glass codeword that matches the result

is the poisoned one. Some will have shorter codewords than others, so the procedure is ended once the result is determined (no redundant tests). They all have equal probability so with $\lceil \log_2 129 \rceil = 8$, there will be $256 - 129 = 127$ symbols with length 7 and $129 - 127 = 2$ with length 8 (look at 5.28 for explanation). Therefore,

$$\mathbb{E}(step) = \frac{127}{129} \times 7 + \frac{2}{129} \times 8 = 7.0155$$

Note also that $H(X) = 7.011$.

Exercise 5.31

Assume that a sequence of symbols from the ensemble X introduced at the beginning of this chapter is compressed using the code C_3. Imagine picking one bit at random from the binary encoded sequence $c = c(x_1)c(x_2)c(x_3)\ldots$. What is the probability that this bit is a 1?

C_3:

a_i	$c(a_i)$	p_i	$h(p_i)$	l_i
a	0	1/2	1.0	1
b	10	1/4	2.0	2
c	110	1/8	3.0	3
d	111	1/8	3.0	3

Solution:

$$\mathbb{E}(\text{number of bits added per symbol}) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = \frac{7}{4}$$

$$\begin{aligned}\mathbb{E}(\text{a bit from code sequence is 1}) &= \frac{\mathbb{E}(\text{number of 1s added per symbol})}{\mathbb{E}(\text{number of bits added per symbol})} \\ &= \frac{\frac{1}{2} \times 0 + \frac{1}{4} \times 1 + \frac{1}{8} \times 2 + \frac{1}{8} \times 3}{\frac{7}{4}} \\ &= \frac{\frac{1}{4} + \frac{1}{4} + \frac{3}{8}}{\frac{7}{4}} = \frac{1}{2}\end{aligned}$$

This makes sense since we know this is an optimal encoding.

Exercise 5.32

How should the binary Huffman encoding scheme be modified to make optimal symbol codes in an encoding alphabet with q symbols? (Also known as 'radix q'.)

Solution:

Each merge decreases by $q-1$. Top-level branch is the most valuable (it has 0-length prefix), but it can end up only merging 2 symbols, so we must avoid that. Hence, we create dummy symbols with $P=0$ so the number of total symbols are $1+k(q-1)$ for the smallest possible k, and then after the encoding discard the dummy symbols. This will eliminate codes with longest length. At most $q-2$ dummy symbols are required so the first merge will always grab all dummy symbols.

Exercise 5.33

Prove that this metacode is incomplete, and explain why this combined code is suboptimal.

Solution:

Let the length of the codeword of x for a child symbol code k be $l_k(x)$. The length of the codeword x for metacode with K child symbol codes will be

$$L(x) = \log_2 K + \min_k l_k(x)$$

Considering the Kraft sum,

$$\begin{aligned}
S &= \sum_x 2^{-L(x)} \\
&= \sum_x 2^{-(\log_2 K + \min_k l_k(x))} \\
&= \frac{1}{K} \sum_x 2^{-\min_k l_k(x)} \\
&= \frac{1}{K} \sum_x \max_k 2^{-l_k(x)} \\
&< \frac{1}{K} \sum_x \sum_k 2^{-l_k(x)} \\
&\leq \frac{1}{K} \sum_k 1 \quad (\because \sum_x 2^{-l_k(x)} \leq 1) \\
&= 1
\end{aligned}$$

so the metacode is incomplete.